UT1 · KOTLIN FUNDAMENTALS · BLOCK 5

Data classes

Practice exercises

These exercises practise section 10 of the notes (**Data classes**). Create them in the package `exercises.classes`, one file per exercise, with the class and `main` in the same file. Do not reuse the names `Contact`, `Grade` or `Point` in other exercises of the same package.

Practice 1 · Contact card

Practises: `data class`, `==` vs `===`, `copy()`, nullable properties.

Write the data class:

```
data class Contact(val name: String, val phone: String, val email: String? = null)
```

1. Create two contacts with the same name and phone. Print one of them, and compare them with `==` and with `===`. Explain the results in a comment.
2. The first contact changes phone and adds an email. Create the updated version with `copy()`, print both versions, and check that the original has not changed.

Expected output

```
Contact(name=Ana, phone=600111222, email=null)
true
false
Contact(name=Ana, phone=699000000, email=ana@mail.com)
Contact(name=Ana, phone=600111222, email=null)
```

Practice 2 · Predict the output

Practises: data class vs normal class, destructuring, `copy()`, `distinct()`.

Without running it, write down what each line prints. Then run it and check your answers. Explain any difference.

```
class Box(val label: String)
data class Point(val x: Int, val y: Int)

fun main() {
    println(Box("A") == Box("A"))
    println(Point(1, 2) == Point(1, 2))

    val p = Point(3, 4)
    val (y, x) = p
    println("x=$x y=$y")

    println(p.copy(y = 0))
    println(listOf(Point(1, 1), Point(1, 1), Point(2, 2)).distinct())
}
```

Practice 3 · Class report

Practises: lists of objects, `map`, `filter`, `maxByOrNull`, `sortedByDescending`, `distinct`.

Write the data class `Grade(val student: String, val subject: String, val mark: Double)` and this list:

```
data class Grade(val student: String, val subject: String, val mark: Double)

val grades = listOf(
    Grade("Laura", "DI", 8.5),
    Grade("Pablo", "DI", 4.0),
    Grade("Marta", "DI", 6.75),
    Grade("Laura", "PMDM", 9.0),
    Grade("Pablo", "PMDM", 5.5)
)
```

Using list functions and lambdas, show:

#	Show…	Expected result
1	The average of all the marks, with two decimals	`6.75`
2	The names of the students with a failed mark	`[Pablo]`
3	The best grade (the whole object)	`Grade(student=Laura, subject=PMDM, mark=9.0)`
4	All the grades sorted from highest to lowest, as "name mark"	`[Laura 9.0, Laura 8.5, Marta 6.75, Pablo 5.5, Pablo 4.0]`
5	Laura's average	`8.75`
6	The list of students without repetitions	`[Laura, Pablo, Marta]`

Practice 4 · Duplicate contacts

Practises: `distinct()` and the `equals()` generated by data classes.

Use the `Contact` class from Practice 1 (do not declare it again: it is in the same package) and this list:

```
val contacts = listOf(
    Contact("Ana", "1"),
    Contact("Luis", "2"),
    Contact("Ana", "1"),
    Contact("Ana", "3")
)
```

Show how many contacts there are, how many **different** contacts there are, and the list of different names.

Expected output

```
Contacts: 4
Different contacts: 3
Different names: [Ana, Luis]
```

TIP · Think about it

Why are `Contact("Ana", "1")` and `Contact("Ana", "3")` different contacts, even though they have the same name?

What would `distinct()` return if `Contact` were a normal class instead of a data class? Try it: remove the word `data` and run it again.